ThreadMoat Website

Administrator's Guide

B2B SaaS platform for industrial AI market intelligence

Version 1.2 — April 2026

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1. System Overview

ThreadMoat Website is a B2B SaaS platform for engineering software and industrial AI market intelligence. It tracks **600+ startups** across the industrial software landscape and presents them through **500+ interactive dashboard visualizations** in **6 languages**. The platform converts visitors into paying subscribers through a tiered access model — from free limited views to enterprise advisory engagements.

What It Does

- **Market intelligence dashboards** — 44+ visualization pages covering market, financial, geographic, network, and advanced analysis
- **Multi-language marketing** — full internationalization across English, French, Spanish, Italian, German, and Portuguese
- **AI-powered analysis** — Claude Sonnet 4.5 generates narrative assessments (Impressions, Conclusions, Beware, Overlooked Opportunities)
- **Custom report builder** — users select companies, toggle sections, include AI narrative, and export PDF/markdown reports
- **Subscription commerce** — Stripe-powered checkout with one-time and annual subscription products

Subscription Tiers

Tier	Price	Access
Recon	Free	Limited dashboard views, masked company names
Analyst	$4,999 (one-time)	Single report purchase with full data access
Strategist	€18,999/yr	Full dashboard access, all reports, AI narratives
Advisory	Custom	Enterprise engagement with direct analyst support

Technology Stack

Layer	Technology
Framework	Next.js 16 (App Router)
Auth	NextAuth
Payments	Stripe (live)
Email	Resend + React Email
Charts	D3, Recharts
Styling	Tailwind CSS, shadcn/ui
i18n	next-intl (6 languages)
AI	Vercel AI SDK + Claude Sonnet 4.5
Database	Neon Postgres
Hosting	Vercel

2. Architecture & File Layout

Directory Structure

```
threadmoat-website/
├── app/
│   ├── [locale]/                  # Public marketing pages with i18n
│   │   ├── page.tsx                 # Home
│   │   ├── pricing/                 # Pricing page
│   │   ├── about/                   # About page
│   │   └── report/                  # Report landing page
│   ├── auth/                      # Authentication flows
│   │   ├── login/
│   │   ├── sign-up/
│   │   ├── forgot-password/
│   │   ├── verify-email/
│   │   └── reset-password/
│   ├── dashboard/                 # 44+ visualization pages
│   │   ├── analytics/
│   │   ├── bar-chart/
│   │   ├── box-plot/
│   │   ├── bubbles/
│   │   ├── buyer-persona/
│   │   ├── candlestick/
│   │   ├── chord/
│   │   ├── compare/
│   │   ├── correlation/
│   │   ├── customer-profile/
│   │   ├── customers/
│   │   ├── distribution/
│   │   ├── explore/
│   │   ├── financial-heatmap/
│   │   ├── growth-momentum/
│   │   ├── heatmap/
│   │   ├── industry-penetration/
│   │   ├── investor-network/
│   │   ├── investor-stats/
│   │   ├── investor-views/
│   │   ├── ip-dependency/
│   │   ├── landscape/
│   │   ├── landscape-intro/
│   │   ├── map/
│   │   ├── marimekko/
│   │   ├── market-momentum/
│   │   ├── maturity-matrix/
│   │   ├── metros/
```

```
│   │   ├── network/
│   │   ├── parallel/
│   │   ├── patterns/
│   │   ├── periodic-table/
│   │   ├── quadrant/
│   │   ├── radar/
│   │   ├── reports/
│   │   ├── sankey/
│   │   ├── settings/
│   │   ├── slope/
│   │   ├── spiral/
│   │   ├── splom/
│   │   ├── sunburst/
│   │   ├── swot/
│   │   ├── tab/
│   │   │   ├── advanced/
│   │   │   ├── financial/
│   │   │   ├── geographic/
│   │   │   ├── market/
│   │   │   └── network/
│   │   ├── tech-independence/
│   │   ├── timeline/
│   │   ├── treemap/
│   │   └── wordcloud/
│   └── api/                        # REST API endpoints
│       ├── auth/
│       ├── companies/
│       ├── funding/
│       ├── investors/
│       ├── profile/
│       │   ├── layout/
│       │   ├── onboarding/
│       │   ├── saved-theses/
│       │   └── thesis/
│       ├── admin/
│       │   ├── coupons/
│       │   └── crm-export/
│       ├── ai/
│       │   └── narrative/
│       └── webhooks/
│           └── stripe/
├── components/                     # React components
│   ├── charts/                       # Chart components (D3, Recharts)
│   ├── checkout/                     # Stripe checkout UI
│   ├── dashboard/                    # Dashboard layout & nav
│   ├── dashboards/                   # Dashboard page wrappers
│   ├── homepage/                     # Marketing page components
│   ├── ui/                           # shadcn/ui primitives
```

```
│   └── widgets/                    # Reusable dashboard widgets
├── contexts/                     # React contexts
│   ├── company-data/
│   ├── filter/
│   ├── layout/
│   ├── plan/
│   ├── scenario/
│   ├── shortlist/
│   └── thesis/
├── lib/                          # Utilities
│   ├── company-data.ts
│   ├── load-companies-server.ts    # Server-side CSV loading
│   ├── stripe.ts
│   ├── subscription.ts
│   ├── tiers.ts
│   ├── db.ts                       # Neon Postgres client
│   ├── email.ts                    # Resend integration
│   ├── rate-limit.ts
│   ├── name-masking.ts             # Non-subscriber name obfuscation
│   └── products.ts
├── data/                         # CSV data files
│   ├── Startups-Grid view.csv
│   ├── Startups-Financial Health.csv
│   ├── Startups-Funding.csv
│   ├── Investors-Grid view.csv
│   ├── heatmap_enrichment.csv
│   └── Startups-Software-only ThreadMoat.csv
├── messages/                     # i18n translation files
│   ├── en.json
│   ├── fr.json
│   ├── es.json
│   ├── it.json
│   ├── de.json
│   └── pt.json
├── scripts/                      # Python data scripts
│   ├── fetch_airtable_csv.py
│   ├── clean_tags.py
│   ├── enrich_op_model.py
│   ├── push_tags_to_airtable.py
│   └── rebuild_tags.py
├── hooks/                        # Custom React hooks
├── migrations/                   # SQL migration files
├── public/                       # Static assets (logos, reports, images
├── next.config.js
├── tailwind.config.ts
└── package.json
```

Data Flow

```
Airtable → scripts/fetch_airtable_csv.py → data/*.csv
                                                ↓
                                lib/load-companies-server.ts (server-side)
                                                ↓
                                contexts/ (client-side React contexts)
                                                ↓
                                components/charts/ → Dashboard pages
                                                ↓
                                Stripe checkout ← /api/webhooks/stripe
```

3. Prerequisites & Environment Setup

Required Software

Software	Version	Purpose
Node.js	20+	Runtime
npm	10+	Package manager
Vercel CLI	Latest	Deployment (optional — git push also works)

Environment Variables

Create a `.env.local` file in the project root:

```
# Authentication
NEXTAUTH_SECRET=...                          # Random secret for session s
NEXTAUTH_URL=http://localhost:3000           # Canonical app URL

# Database
NEON_DATABASE_URL=postgres://...             # Neon Postgres connection st

# Stripe (live keys)
STRIPE_SECRET_KEY=sk_live_...                # Stripe secret key
STRIPE_WEBHOOK_SECRET=whsec_...              # Stripe webhook signing secr
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=pk_live_  # Stripe publishable key (cli

# Email
RESEND_API_KEY=re_...                        # Resend transactional email

# Data source
AIRTABLE_TOKEN=pat...                        # Airtable Personal Access To
AIRTABLE_BASE_ID=app...                      # Airtable base ID

# AI
ANTHROPIC_API_KEY=sk-ant-...                 # Claude API key for AI narra
```

Security: Never commit `.env.local` to git. The `.gitignore` already excludes it. For production, set all variables in the Vercel dashboard.

4. Local Development

First-Time Setup

```
# Install dependencies
npm install

# Start development server
npm run dev
```

The app will be available at http://localhost:3000.

Common Commands

Command	Purpose
`npm run dev`	Start development server (localhost:3000)
`npm run build`	Production build — catches TypeScript and build errors
`npm run lint`	Run ESLint across the project
`npm start`	Start production server (after build)

> **Tip:** Always run `npm run build` before pushing to catch TypeScript errors that the dev server may not surface.

5. Data Pipeline

The website reads startup and investor data from CSV files stored in the `data/` directory. These CSVs are exported from the Airtable master database and loaded server-side via `lib/load-companies-server.ts`.

CSV Data Files

File	Contents	Rows (approx)
`Startups-Grid view.csv`	Main startup data: name, description, URL, HQ, founded, employees, category, discipline, lifecycle phase	740
`Startups-Financial Health.csv`	Financial metrics: revenue estimates, burn rate, runway, valuation	740
`Startups-Funding.csv`	Funding history: rounds, amounts, investors, dates	740
`Investors-Grid view.csv`	Investor profiles: name, type, fund size, portfolio companies, focus areas	200+
`heatmap_enrichment.csv`	Derived heatmap columns from the Agentic Updates project	600
`Startups-Software-only ThreadMoat.csv`	Filtered view: software-only startups used for primary dashboard display	600

Heatmap Enrichment CSV

The `heatmap_enrichment.csv` is produced by the ThreadMoat Agentic Updates project. It contains 600 rows and 7 columns:

Column	Description
Name	Company name (join key)
Company URL	Primary URL
TIS	Technology Independence Score (0–100)
Ecosystem Dependencies	Vendor ecosystems the company depends on
Growth Momentum Tier	Accelerating / Steady / Decelerating / Early / Stalled
Buyer Persona	Primary buyer role (Design Engineering, Manufacturing/OT, etc.)
Data Completeness	N/4 quality flag for TIS reliability

Refreshing Data

```
# Pull fresh CSVs from Airtable
python scripts/fetch_airtable_csv.py

# Then redeploy (git push or vercel deploy)
```

Note: The `fetch_airtable_csv.py` script requires `AIRTABLE_TOKEN` and `AIRTABLE_BASE_ID` environment variables. The heatmap_enrichment.csv must be copied separately from the Agentic Updates project output.

6. Authentication & Access Control

Authentication is handled by NextAuth backed by Neon Postgres for session and user storage.

Auth Flows

Route	Purpose
`/auth/sign-up`	New user registration with email verification
`/auth/login`	Email/password login
`/auth/forgot-password`	Password reset request (sends email via Resend)
`/auth/reset-password`	Password reset form (from email link)
`/auth/verify-email`	Email verification callback

Tier Access Control

Tier	Price	Dashboard Access	Company Names	AI Narrative	Reports
Recon (free)	$0	Limited views	Masked	No	No
Analyst	$4,999	Full (one-time)	Visible	Yes	1 report
Strategist	€18,999/yr	Full (subscription)	Visible	Yes	Unlimited
Advisory	Custom	Full + priority	Visible	Yes	Unlimited

Name Masking

Non-subscribers see obfuscated company names via `lib/name-masking.ts`. This masks company names, URLs, and other identifying details in dashboard views and API responses. Full data is only revealed after Stripe checkout completes and the user's tier is upgraded.

Coupon System

Partners and early adopters can receive coupons via the admin API (`/api/admin/coupons`). Coupons map to specific tier upgrades and can be one-time or multi-use.

Onboarding

New subscribers are guided through an onboarding wizard that collects their role, company type, and investment focus areas. This data powers the thesis context and personalized dashboard defaults.

7. Internationalization

The website uses **next-intl** for internationalization across 6 languages.

Supported Languages

Code	Language	URL Prefix
en	English	`/` (default, no prefix)
fr	French	`/fr`
es	Spanish	`/es`
it	Italian	`/it`
de	German	`/de`
pt	Portuguese	`/pt`

Configuration

- **Locale prefix:** `as-needed` — English pages have no prefix (`/pricing`), other languages use their code (`/fr/pricing`)
- **Translation files:** stored in `messages/` as JSON files (one per language)
- **Language switcher:** uses flag emojis in the navigation bar

> **Brand names are never translated.** ThreadMoat, Digital Thread, Analyst, and Strategist remain in English across all locales. These terms are used as product/tier names and must be consistent.

Adding a Translation Key

1. Add the key to `messages/en.json`
2. Add translations to all other locale files (`fr.json` , `es.json` , `it.json` , `de.json` , `pt.json`)
3. Use `useTranslations('namespace')` in the component to access it

8. Dashboard Visualizations

The dashboard contains **44+ visualization pages** organized into tab groups. All charts respond to the global filter context and thesis context for filtered views by investment thesis.

Tab Organization

Tab	Route	Key Visualizations
Market	`/dashboard/tab/market`	Periodic Table, Landscape, Quadrant, SWOT, Distribution, Patterns
Financial	`/dashboard/tab/financial`	Financial Heatmap, Candlestick, Bar Chart, Box Plot, Slope
Geographic	`/dashboard/tab/geographic`	3D Globe Map, Metros, Treemap
Network	`/dashboard/tab/network`	Investor Network, Chord, Sankey, Sunburst
Advanced	`/dashboard/tab/advanced`	Tech Independence, Buyer Persona, Marimekko, Parallel, Spiral, SPLOM, Correlation

Key Visualizations

Chart	Route	Description
Periodic Table	`/dashboard/periodic-table`	Interactive periodic-table–style grid of all startups by category
3D Globe Map	`/dashboard/map`	Globe visualization of startup HQ locations worldwide
Investor Network	`/dashboard/investor-network`	Force-directed graph showing co-investment relationships
Financial Heatmap	`/dashboard/financial-heatmap`	Heatmap of financial health metrics across startups
Market Momentum	`/dashboard/market-momentum`	Heatmap of growth momentum tiers across investment lists
Industry Penetration	`/dashboard/industry-penetration`	Heatmap showing startup penetration across industrial verticals
Customer Profile	`/dashboard/customer-profile`	Heatmap of target customer profiles by buyer persona
Tech Independence	`/dashboard/tech-independence`	TIS scores and vendor ecosystem dependency analysis
Buyer Persona	`/dashboard/buyer-persona`	Distribution of primary buyer roles across startups
Sankey	`/dashboard/sankey`	Funding flow diagram from investors to startups by category
Marimekko	`/dashboard/marimekko`	Proportional market share visualization by category and funding
Wordcloud	`/dashboard/wordcloud`	Tag frequency visualization from startup descriptions

Contexts

- **Filter context** (`contexts/filter/`) — global filters (category, funding range, geography, lifecycle phase) that apply across all chart pages
- **Thesis context** (`contexts/thesis/`) — investment thesis–based filtering: users define a thesis and all views narrow to matching companies
- **Company data context** (`contexts/company-data/`) — loaded once from server, shared across all dashboard components

- **Shortlist context** (`contexts/shortlist/`) — user's saved list of companies for report generation

9. AI Narrative Engine

The AI narrative endpoint generates analytical assessments of companies using Claude Sonnet 4.5 via the Vercel AI SDK.

Endpoint

```
POST /api/ai/narrative
```

Output Sections

Section	Purpose
Impressions	First-look assessment of the company's market position and technology
Conclusions	Investment-grade conclusions based on financial and competitive data
Beware	Risk factors, red flags, and competitive threats to monitor
Overlooked Opportunities	Underappreciated strengths, adjacent markets, and strategic angles

The endpoint receives a company's full data profile (from the loaded CSVs) and generates structured narrative sections. It is used by the custom report builder and is gated behind Analyst tier and above.

> **Cost:** Each AI narrative call consumes Claude API tokens via the `ANTHROPIC_API_KEY`. Monitor usage in the Anthropic dashboard.

10. Custom Report Builder

The report builder allows subscribers to create custom market intelligence reports.

Workflow

1. **Select companies** — choose from shortlist or search the full database
2. **Toggle report sections** — include/exclude company overview, financials, competitive analysis, investor landscape

3. **Include AI narrative** — optionally generate Claude-powered analysis for each selected company
4. **Capture chart screenshots** — embed relevant dashboard visualizations in the report
5. **Export** — generate PDF or markdown output

Reports are available to Analyst (one report per purchase), Strategist (unlimited), and Advisory tier users.

11. Stripe Integration

The website uses **Stripe in live mode** for payment processing.

Products

Product	Type	Price
Analyst Report	One-time payment	$4,999
Strategist Subscription	Annual recurring	€18,999/yr

Integration Points

Component	Purpose
`lib/stripe.ts`	Stripe SDK initialization and helpers
`lib/subscription.ts`	Subscription status checking and tier resolution
`lib/products.ts`	Product/price ID mappings
`components/checkout/`	Checkout UI components
`/api/webhooks/stripe`	Webhook handler for payment events
`/api/admin/coupons`	Coupon management for partners

Webhook Events Handled

- `checkout.session.completed` — upgrade user tier after successful payment
- `customer.subscription.updated` — handle plan changes
- `customer.subscription.deleted` — downgrade user to Recon on cancellation
- `invoice.payment_failed` — flag account for follow-up

> **Critical:** The `STRIPE_WEBHOOK_SECRET` must match the webhook endpoint configured in the Stripe dashboard. A mismatch silently drops all payment events, breaking tier upgrades.

12. Deployment

The website is hosted on **Vercel** with automatic deployments on git push.

Deployment Methods

Method	Command	When
Git push	`git push origin main`	Standard — push to main triggers auto-deploy
Vercel CLI	`vercel --prod`	Manual deploy from local machine
Preview	Push to any non-main branch	Preview deployment with unique URL

Environment Variables

All environment variables from Section 3 must be configured in the **Vercel dashboard** under Project Settings → Environment Variables. Set them for Production, Preview, and Development environments as appropriate.

Domain

Production domain: **threadmoat.com**

13. Data Update Workflow

To refresh the startup data displayed on the website, follow this step-by-step process:

Step 1: Pull Fresh CSVs from Airtable

```
# From the threadmoat-website project root
python scripts/fetch_airtable_csv.py
```

This downloads the latest Airtable views into `data/`.

Step 2: Copy Heatmap Enrichment Data

```
# Copy from the Agentic Updates project output
cp ../ThreadMoat-Agentic-Updates/intake_output/heatmap_enrichment.csv data
```

Step 3: Commit Changes

```
git add data/
git commit -m "data: refresh CSVs from Airtable ($(date +%Y-%m-%d))"
```

Step 4: Deploy

```
git push origin main
```

Vercel auto-deploys within ~60 seconds of push.

> **Frequency:** Data updates are typically done weekly, coinciding with the Agentic Updates weekly sweep. The heatmap enrichment CSV is regenerated less frequently (bi-weekly or as needed).

14. Troubleshooting

Common Issues

Build Failures

Run `npm run build` locally to reproduce the error. Most build failures are TypeScript errors — the error output will show the file, line, and type mismatch. Common causes:

- Missing props on a component after a refactor
- Importing a server-only module in a client component (`"use client"`)
- Unused variables or imports (strict lint rules)

Stripe Webhook Not Firing

Symptoms: users complete checkout but their tier isn't upgraded.

- Verify `STRIPE_WEBHOOK_SECRET` matches the secret in Stripe Dashboard → Webhooks
- Check the webhook endpoint URL is `https://threadmoat.com/api/webhooks/stripe`

- Review Stripe Dashboard → Webhooks → Recent deliveries for error codes
- For local testing, use `stripe listen --forward-to localhost:3000/api/webhooks/stripe`

CSV Parse Errors

Symptoms: dashboard shows no data or crashes on load.

- Check CSV files use UTF-8 encoding and comma delimiters
- Open the CSV in a text editor — look for unescaped commas in text fields or broken line endings
- Verify column headers match what `lib/load-companies-server.ts` expects
- Re-export from Airtable if the file is corrupted

i18n Missing Key Warnings

Symptoms: console warnings like `Missing message: namespace.key for locale "fr"`.

- Check that the key exists in all 6 files under `messages/`
- Keys must match exactly (case-sensitive, same nesting structure)
- Add the missing translation — next-intl falls back to the default locale (English) but logs a warning

AI Narrative Errors

Symptoms: report builder shows "Failed to generate narrative" error.

- Verify `ANTHROPIC_API_KEY` is set and valid
- Check the Anthropic dashboard for rate limit or billing issues
- The narrative endpoint has a 30-second timeout — very large company profiles may hit it

Name Masking Showing for Subscribers

Symptoms: paying users still see masked company names.

- Check that the Stripe webhook processed the payment event (see webhook section)
- Verify the user's tier in the database: `SELECT tier FROM users WHERE email = '...'`
- Clear the user's session (sign out / sign in) to refresh the tier from the database

15. Quick Reference

Commands

Task	Command
Install dependencies	`npm install`
Start dev server	`npm run dev`
Production build	`npm run build`
Lint	`npm run lint`
Pull fresh CSVs	`python scripts/fetch_airtable_csv.py`
Deploy to production	`git push origin main`
Manual deploy	`vercel --prod`
Stripe webhook testing	`stripe listen --forward-to localhost:3000/api/webhooks/stripe`

Key URLs

URL	Purpose
`https://threadmoat.com`	Production website
`http://localhost:3000`	Local development
`https://vercel.com/dashboard`	Vercel deployment dashboard
`https://dashboard.stripe.com`	Stripe payment dashboard
`https://neon.tech/console`	Neon Postgres console
`https://resend.com/dashboard`	Resend email dashboard
`https://console.anthropic.com`	Anthropic API usage
`https://airtable.com`	Master startup database

Key Files

File	Purpose
`lib/load-companies-server.ts`	Server-side CSV loading — entry point for all startup data
`lib/name-masking.ts`	Company name obfuscation for non-subscribers
`lib/tiers.ts`	Tier definitions and access control logic
`lib/stripe.ts`	Stripe SDK setup and checkout session creation
`contexts/filter/`	Global filter state shared across all dashboard pages
`contexts/thesis/`	Investment thesis context for filtered views
`messages/en.json`	Source-of-truth translation file (English)
`data/heatmap_enrichment.csv`	Heatmap columns from Agentic Updates project